KICKBOTS ARENA

PRESENTED BY OHMIES

GROUP - 12

ITM

Team Members

- 245031B - P.D.V.D.Gunasekara
- 245056F - K.Lathursan
- 245124M - S.M.M.A.B. Madugalla
- 245057J - B.L.N.N.Liyanage
- 245060L - S.M.K.Madhushani

OUTLINE

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Introduction

KickBots Arena is an innovative 2-player robotic football system . Each player controls a striker robot and its kicker for attacking and a goalkeeper robot for defense. A Web application acts as the main control and display system.

Aim & Objectives

Aim

To design and develop a 2-player robotic football arena where each player controls a striker and goalkeeper robot using wireless communication and an interactive web interface, enabling a real-time, competitive and automated gaming experience.

Objectives

- Build striker and goalkeeper robots with precise movement
- Implement wireless control using joysticks and nRF24L01 modules
- Create a web-based interface to display timer, match status, and live score updates
- Detect goals automatically using IR sensor modules
- Control goalkeeper movement on a linear rail system using DC motors and limit switches
- Ensure all microcontrollers (Arduino, ESP32) communicate reliably in real-time
- Develop a reset and match-restart functionality with scoring history
- Enable a smooth multiplayer experience with minimal delay

Problem &Solutions

Problem

Traditional tabletop and arcade football games lack realistic control, automation, and accurate scoring. Player actions like shooting or goalkeeping are limited, and scoring is manual and error-prone. Most robotic games are single-player, offering no real-time competition or engaging, automated gameplay.

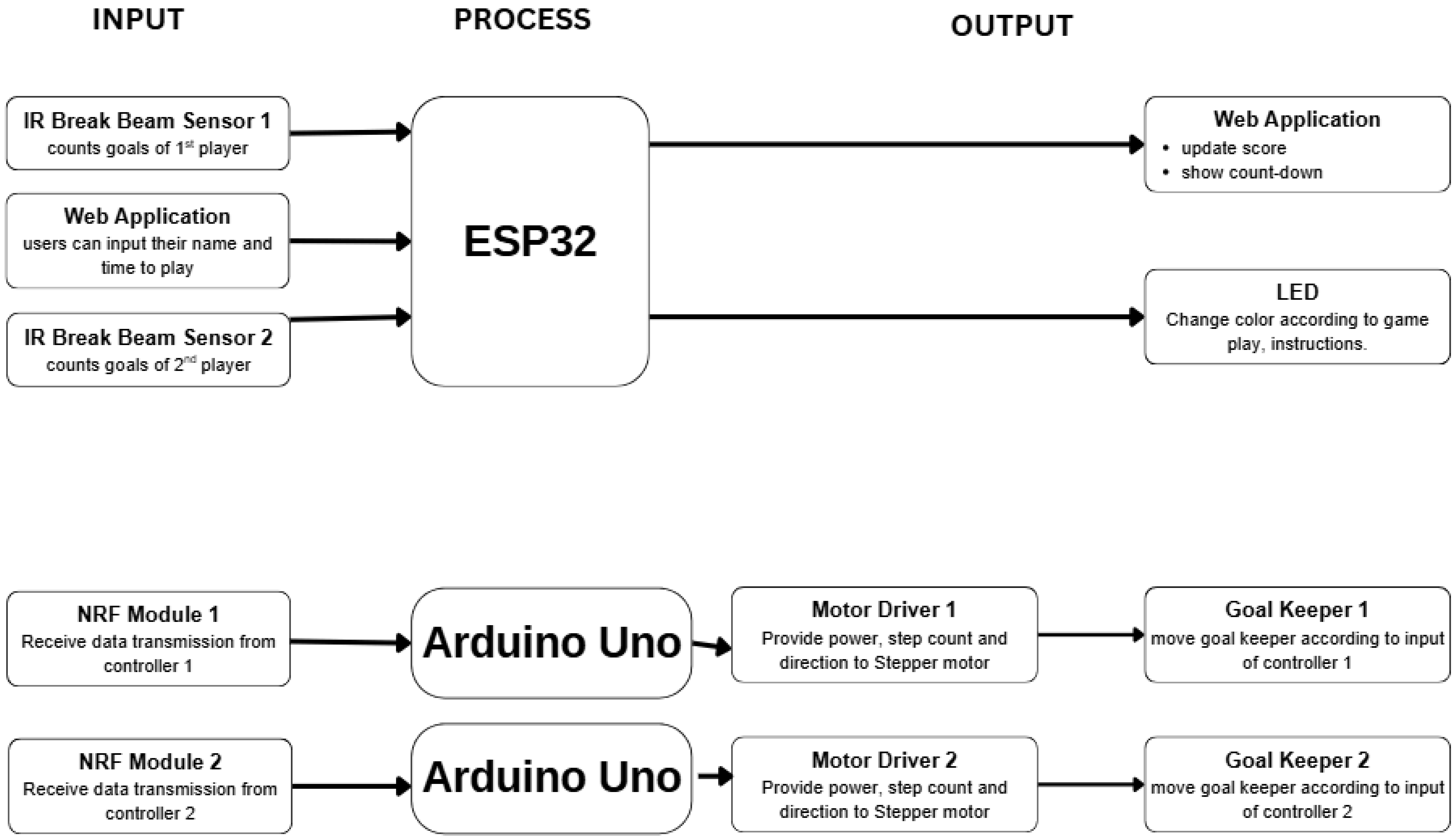
Solution and features

KickBots Arena is a 2-player robotic football game with full automation and wireless control.

- Each player controls two robots: a striker and a goalkeeper (on a rail).
- IR sensors detect goals, and scores update instantly on a web app.
- The web interface handles match time, controls, and winner display.
- Wireless modules ensure smooth and responsive gameplay.

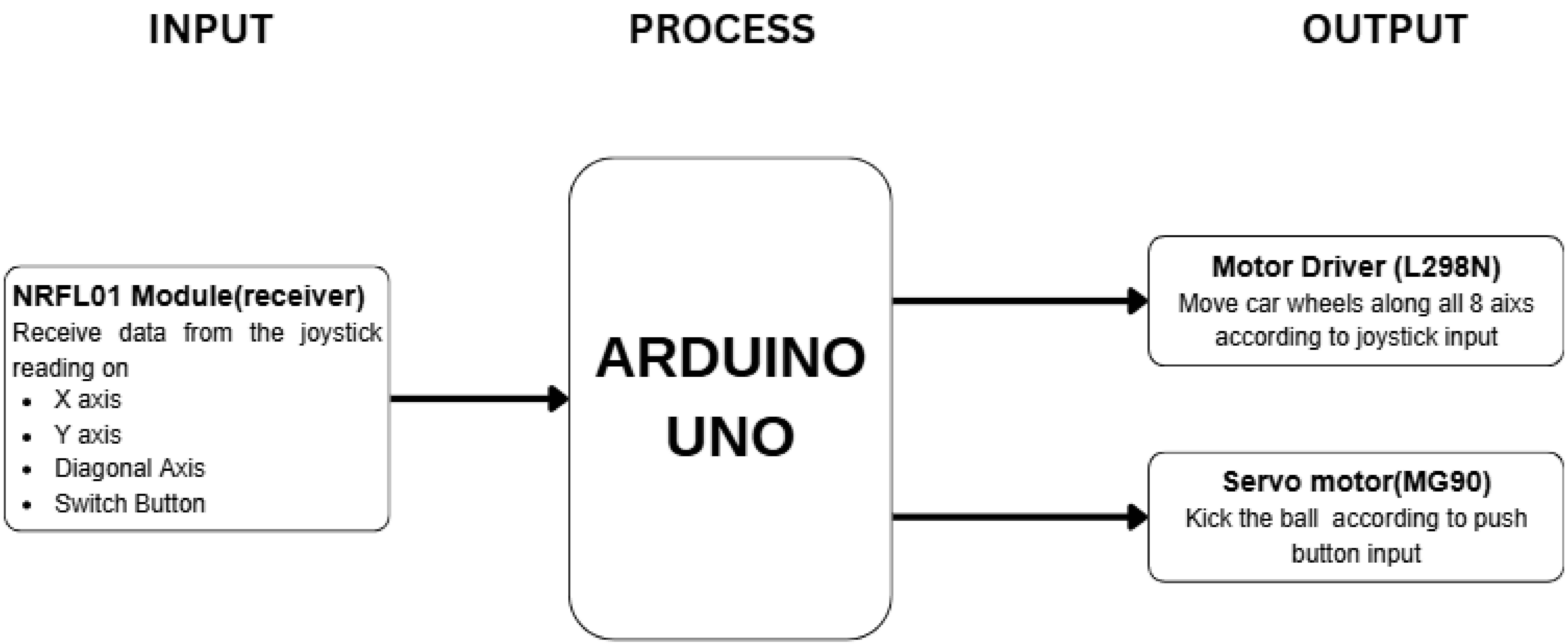
Block diagram

01. ARENA



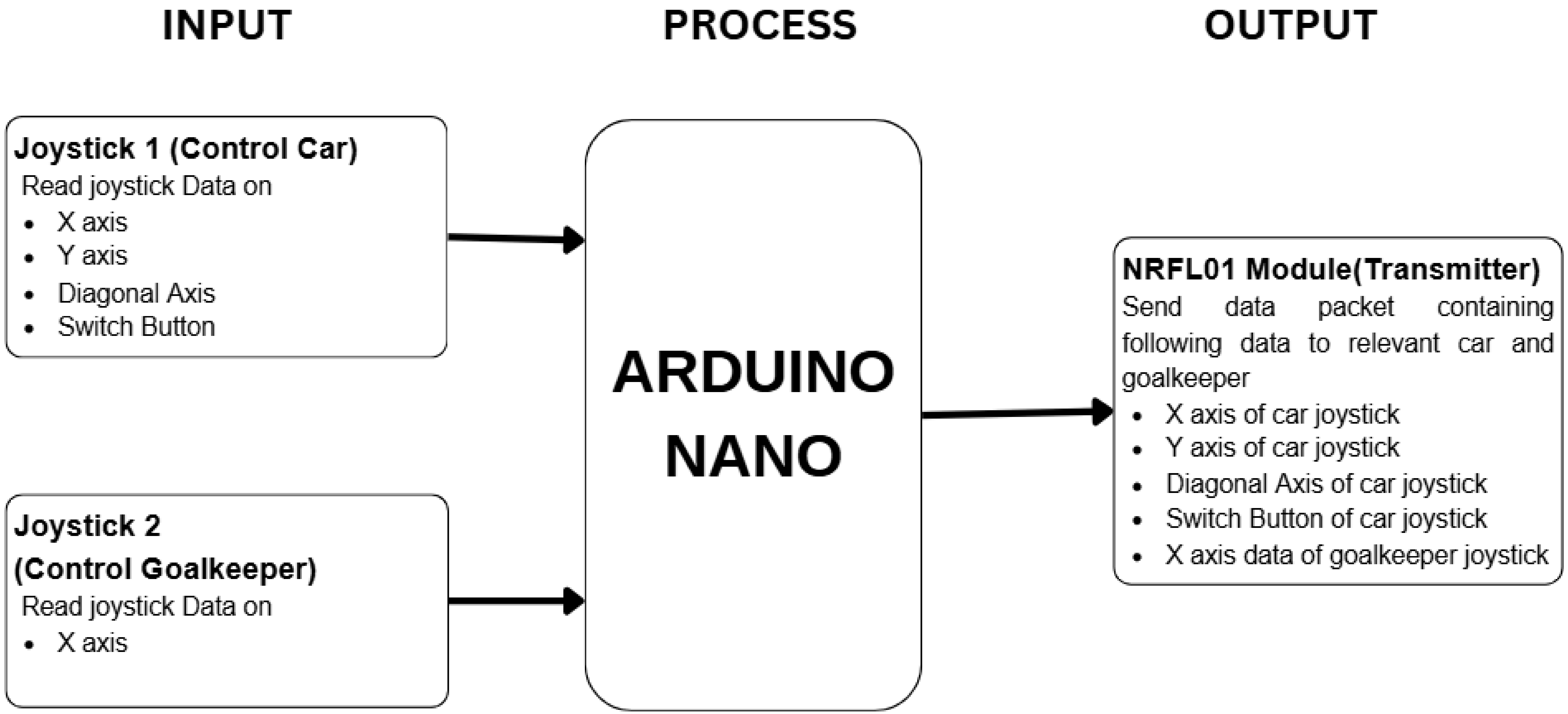
Block diagram

02. CAR



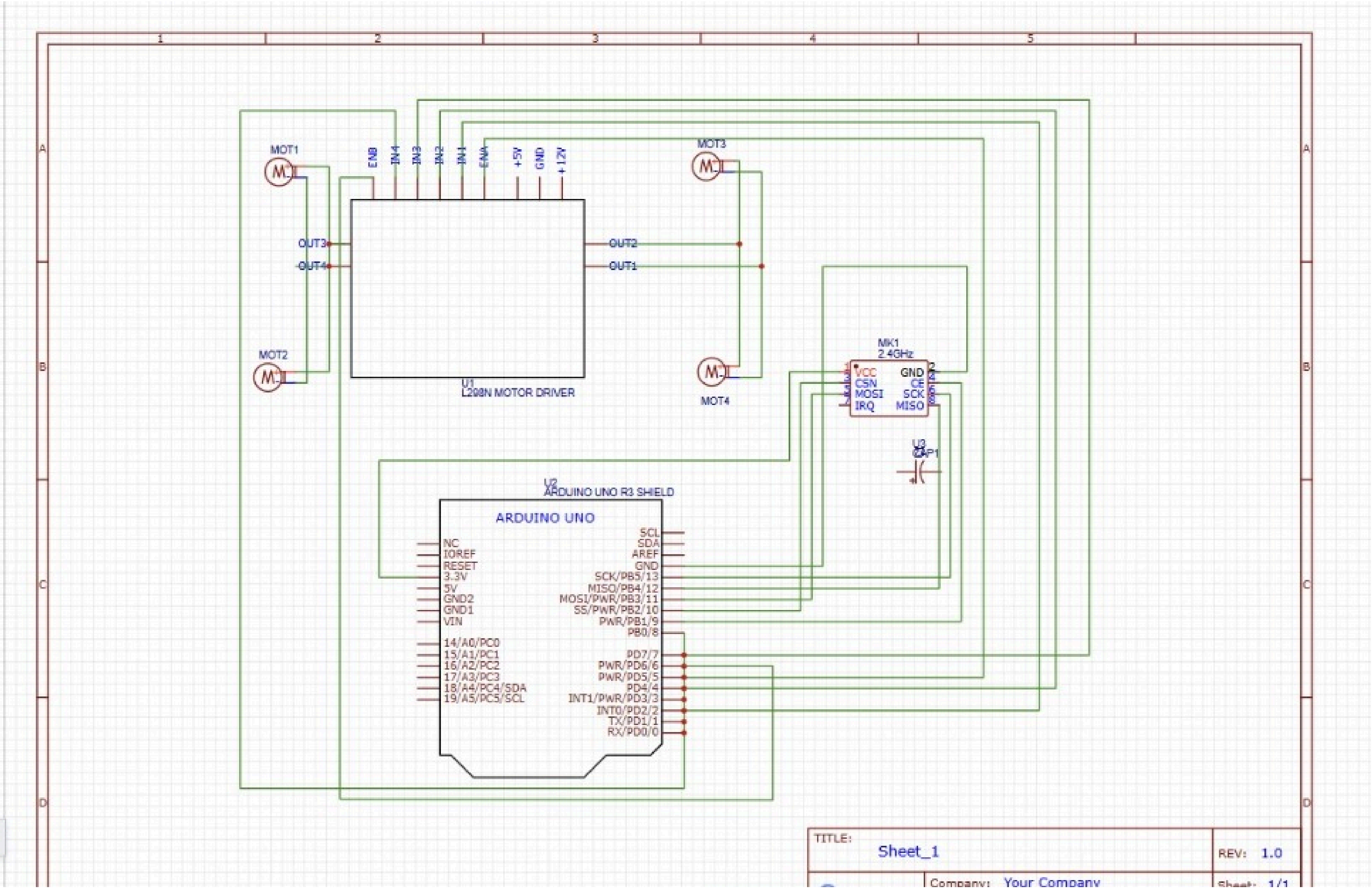
Block diagram

03. CONTROLLER



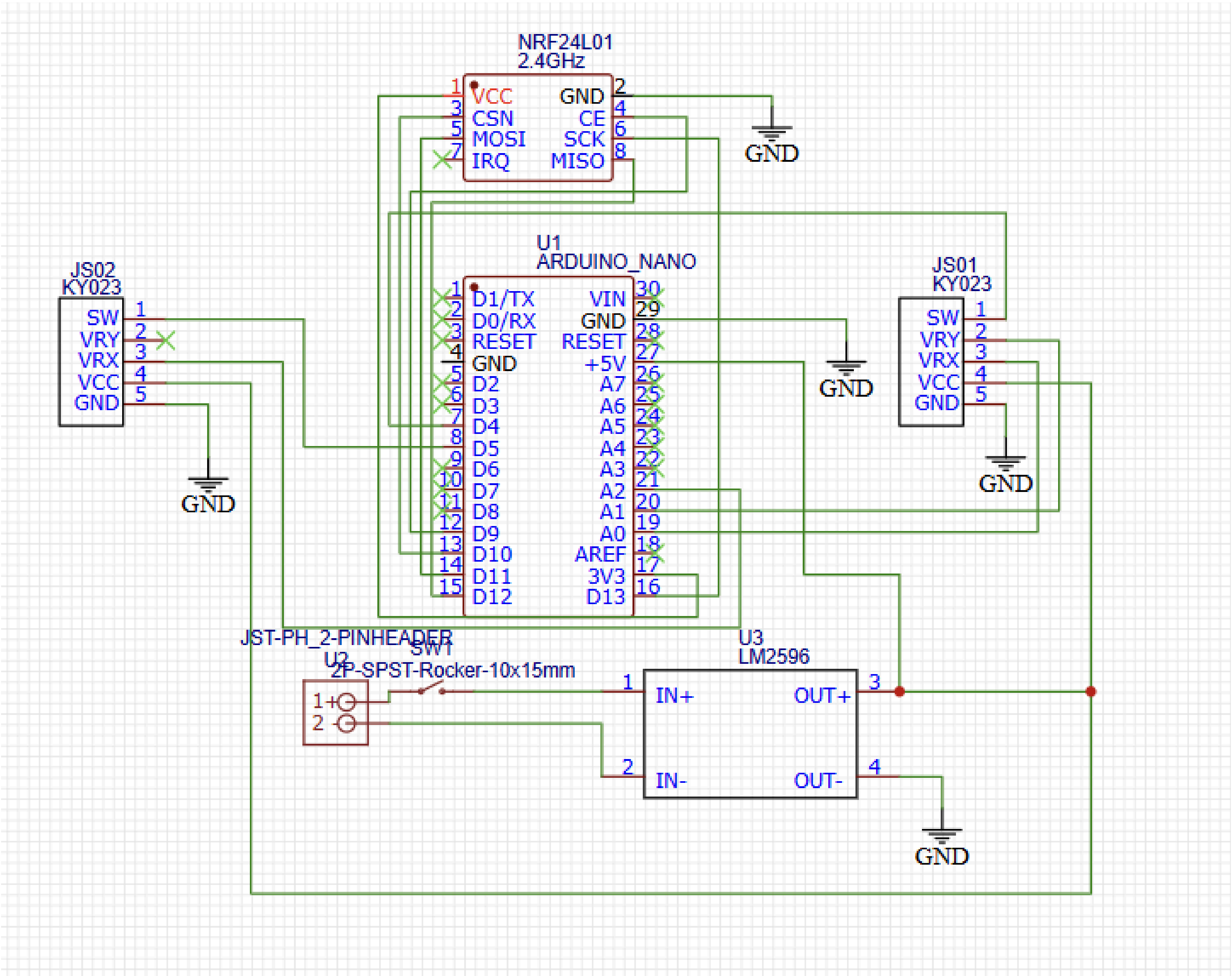
Circuit Diagram

1. Car



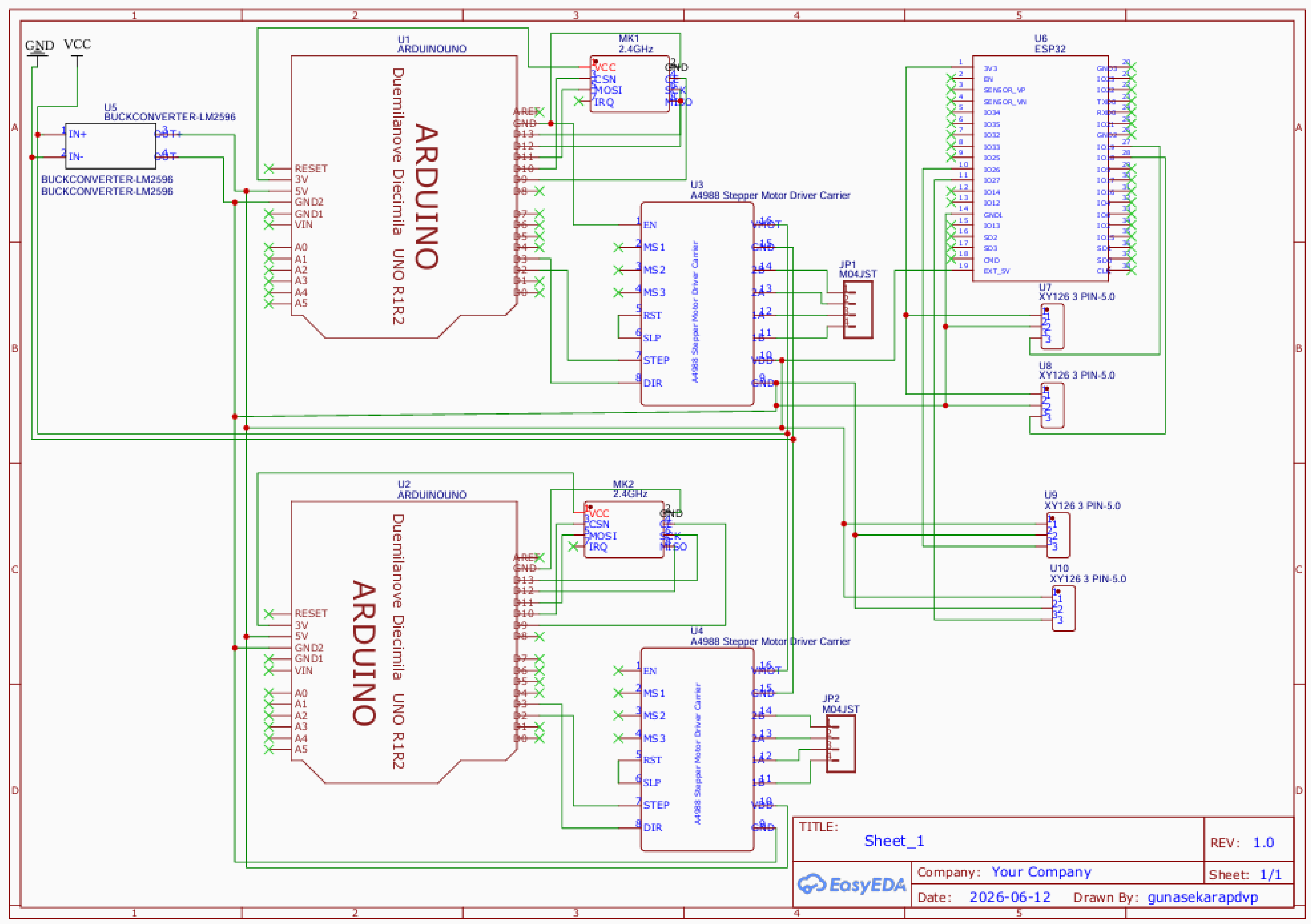
Circuit Diagram

2. Controller



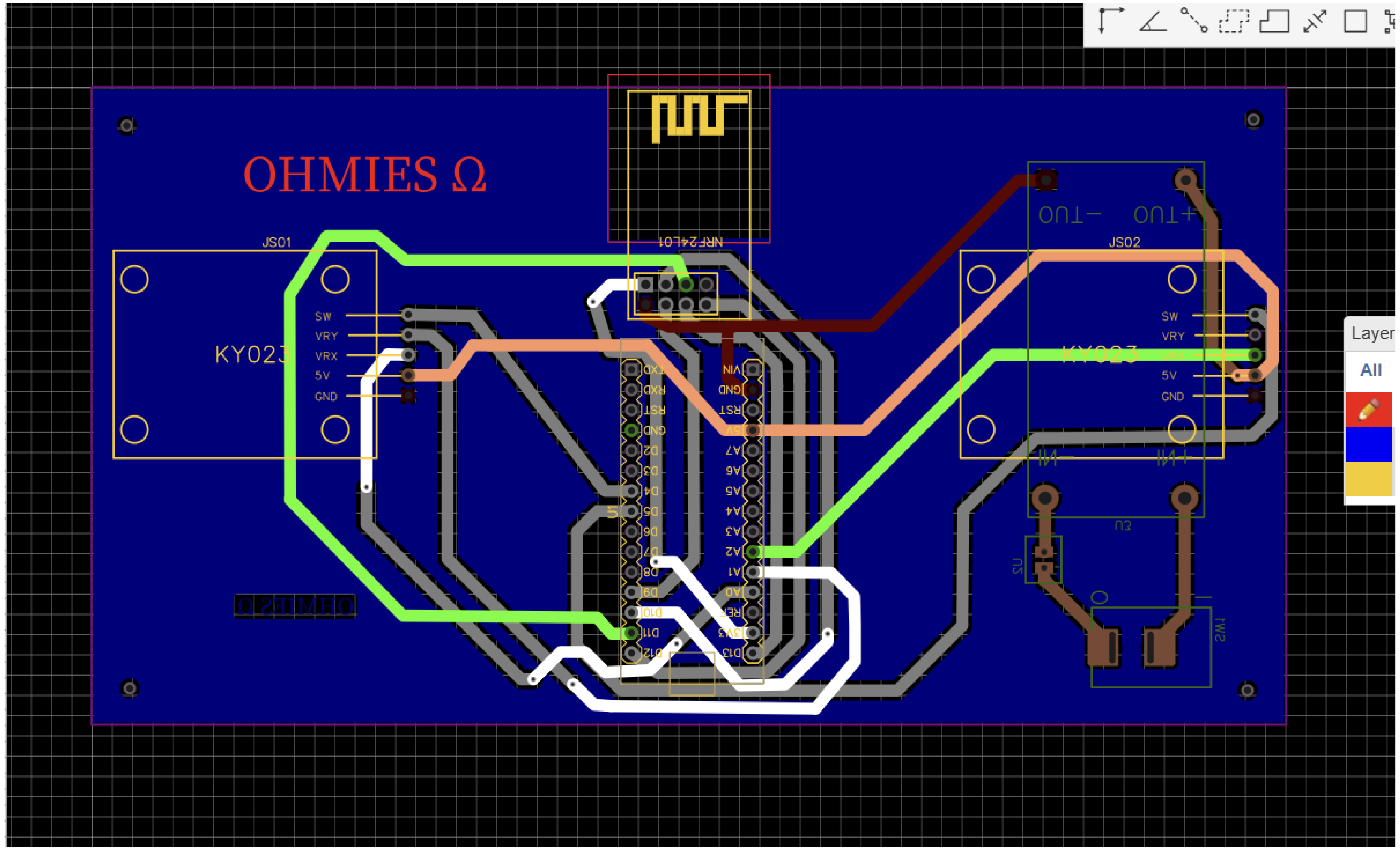
Circuit Diagram

3. Arena



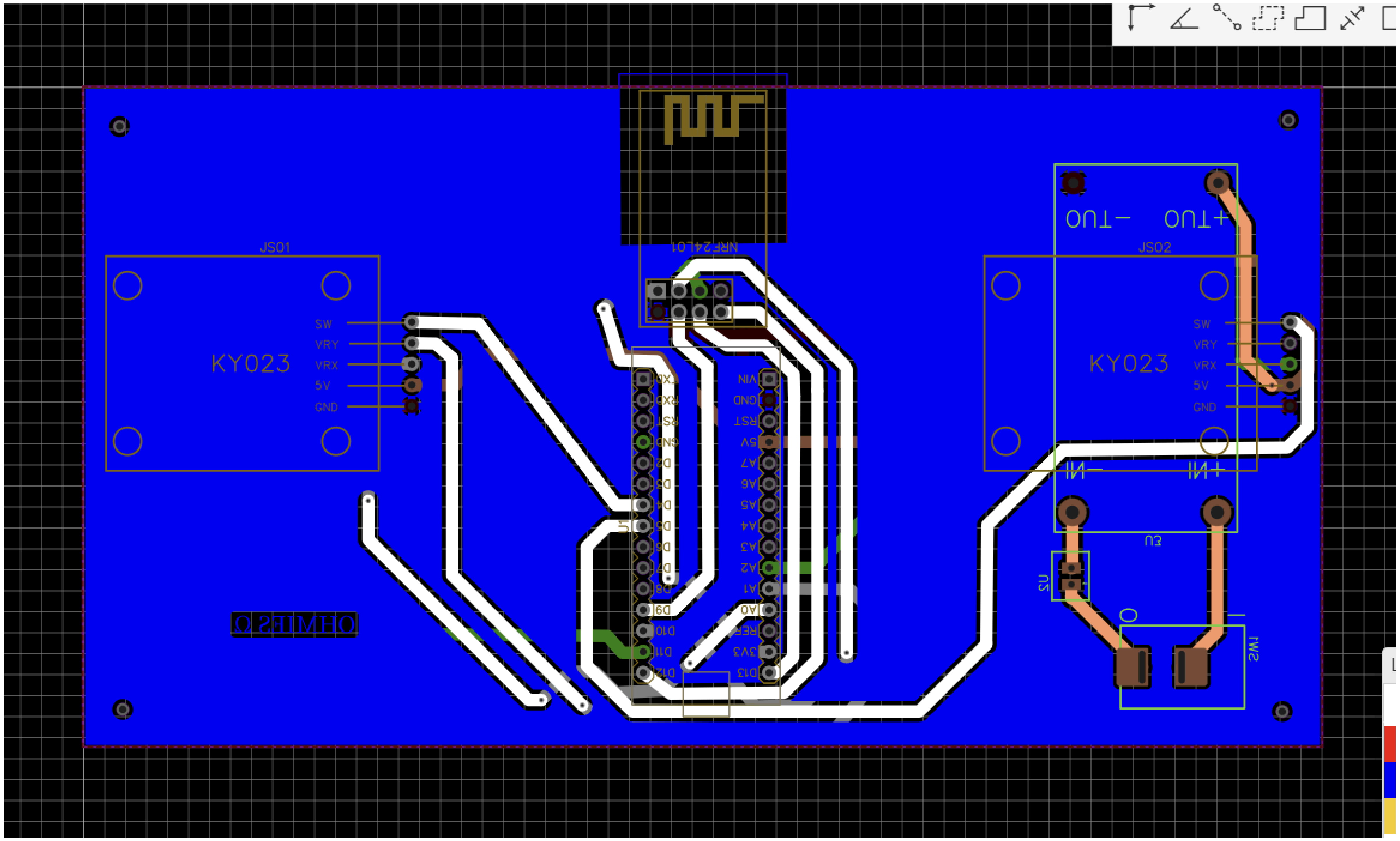
PDB Design

For controller-Top Layer



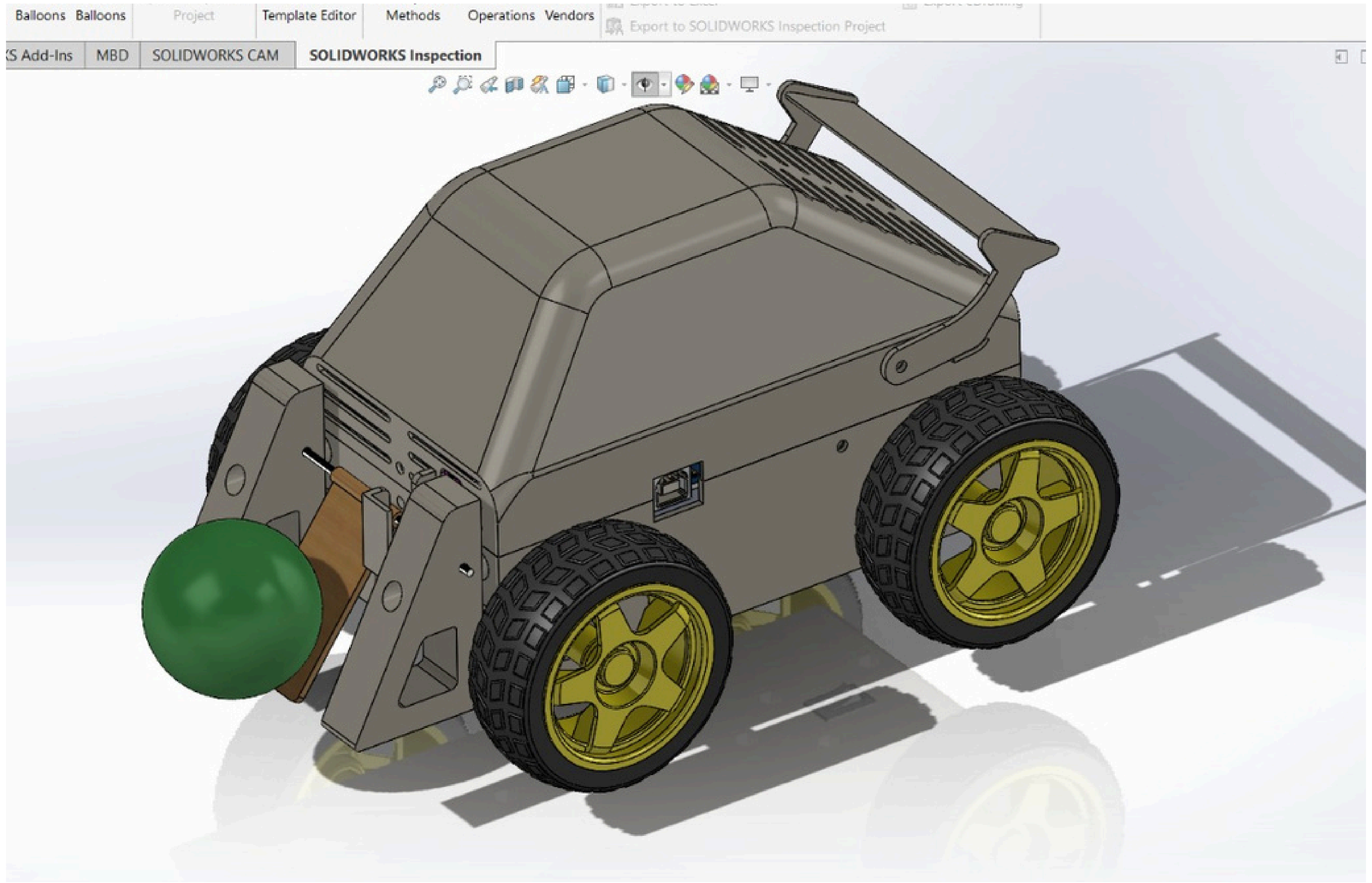
PDB Design

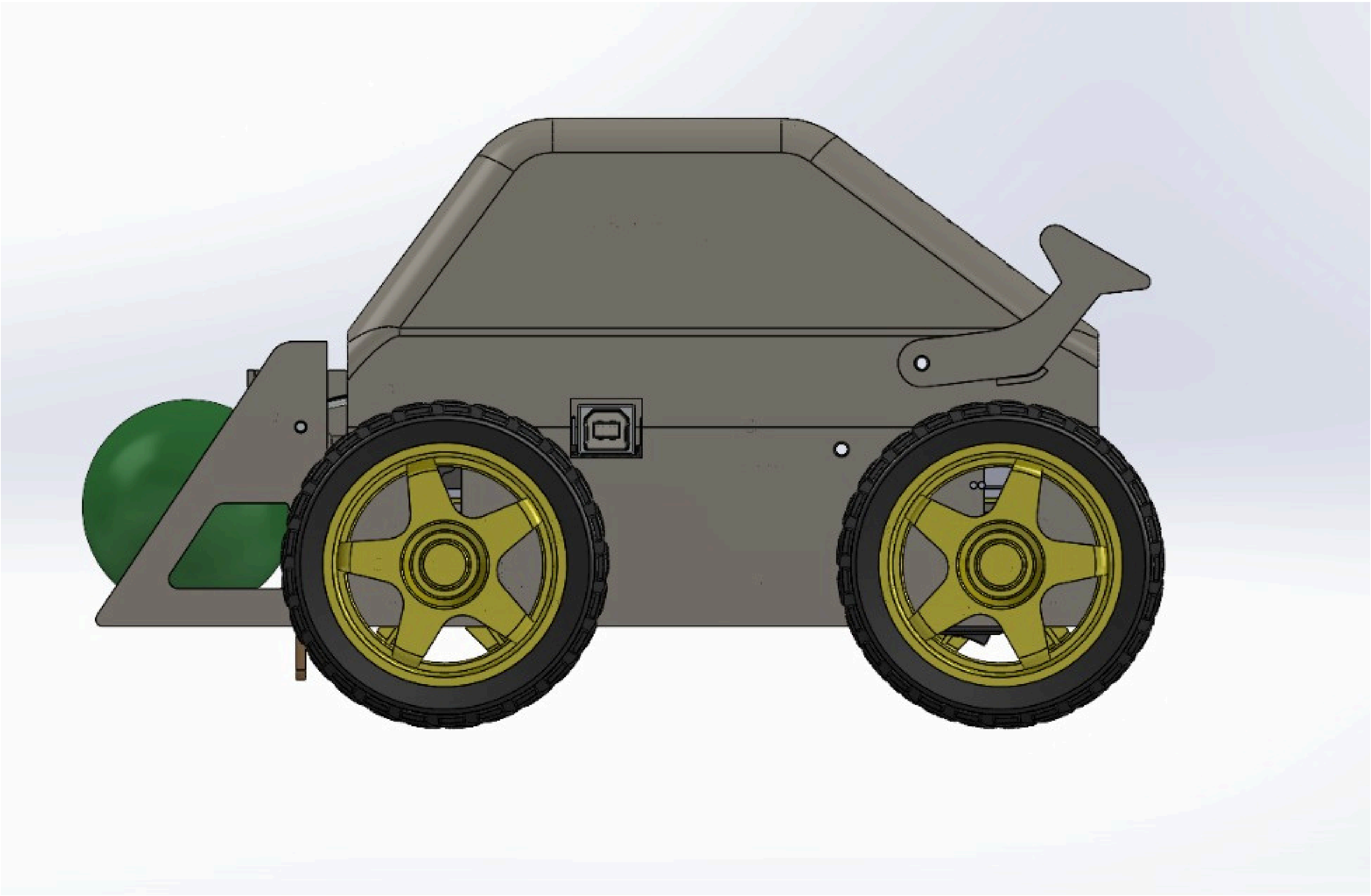
For controller-Bottom layer



3D Model

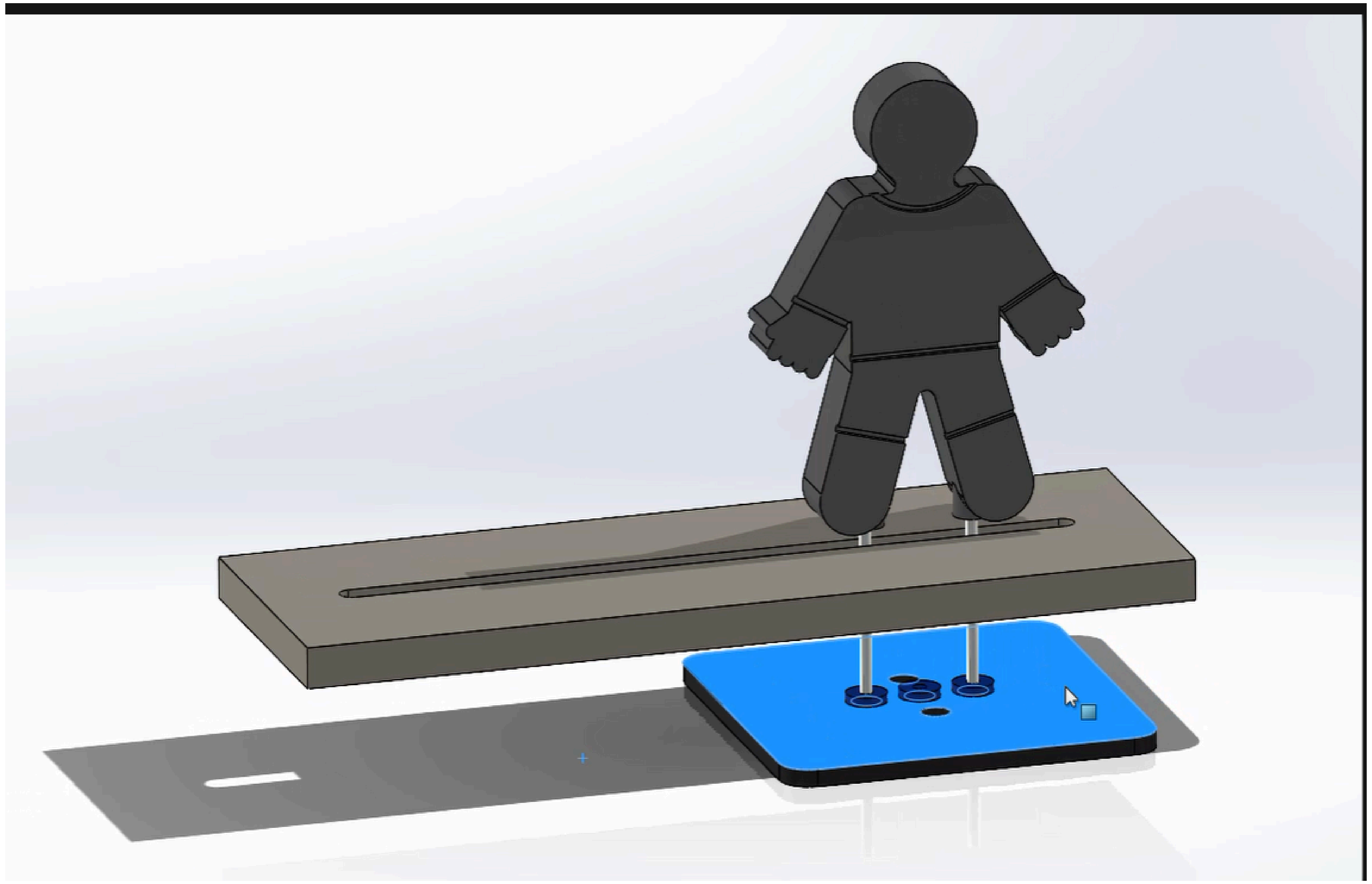
Car

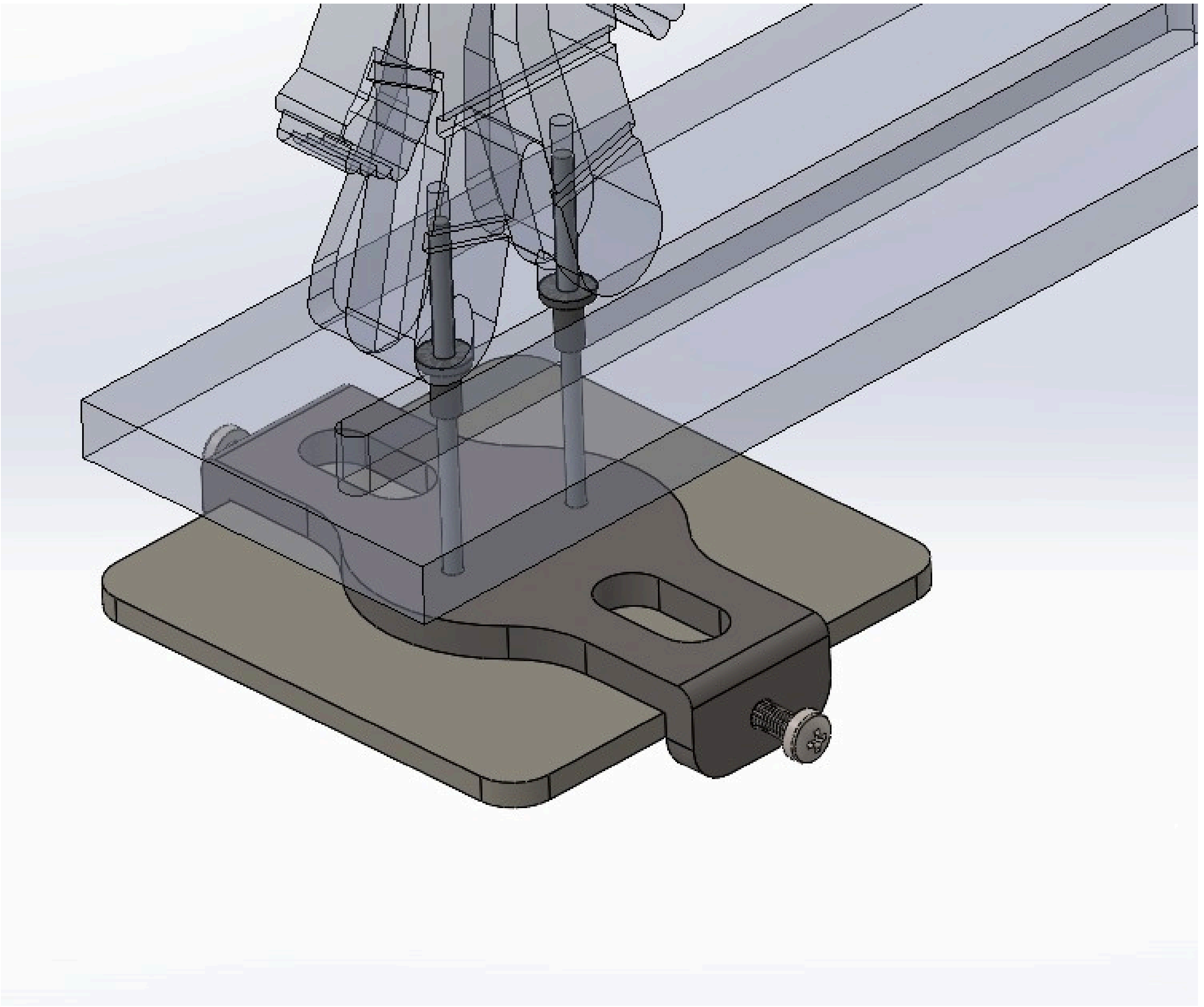

Balloons
Balloons
Project
Template Editor
Methods
Operations
Vendors
Export to SOLIDWORKS Inspection Project
S Add-Ins
MBD
SOLIDWORKS CAM
SOLIDWORKS Inspection



3D Model

Goal Keeper





Hardware Components

STRIKER ROBOTS

- Arduino Uno
- Motor Driver (L298N)
- 4× BO gear motors (150–200 RPM)
- Mini Servor (Kicker)
- nRF24L01 wireless receiver
- 2× 18650 Li-ion cells (7.4 V total)
- Acrylic or 3D printed chassis (~12–15 cm × 10 cm × 7 cm)

JOYSTICK CONTROLLER

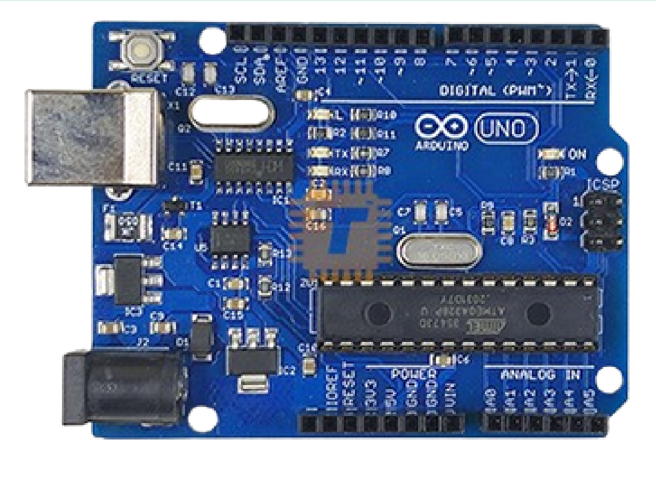
- Arduino Nano
- 2× Joysticks (for striker & goalkeeper)
- Push button
- nRF24L01 wireless transmitter
- 1× 18650 Li-ion battery
- Boost converter (5V output)

ARENA

- ESP32 WROOM32
- NEMA 17 Stepper Motor
- GT2 timing belt or linear slide system (20 cm travel range)
- A4988 motor driver
- nRF24L01 wireless receiver
- 6–9 V DC power supply (shared)

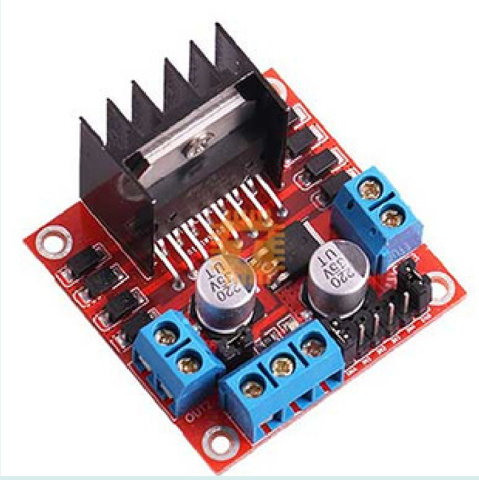
COMPONENTS USED

1.Arduino Uno



- Operating Voltage: 5V
- Input Voltage (Recommended): 7-12V
- Digital I/O Pins: 14 (6 provide PWM output)
- Analog Input Pins: 6
- Flash Memory: 32 KB (ATmega328P)

2.L298N Motor Driver Module

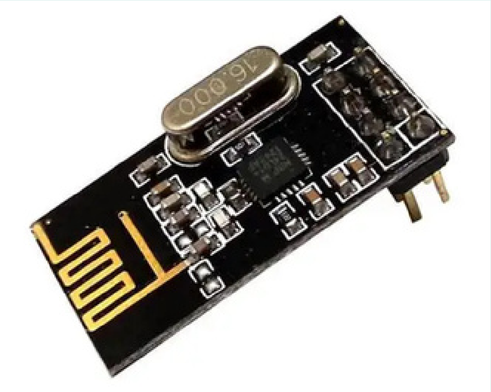


- Operating Voltage: 5V–35V
- Peak Current: 2A per channel
- Logical Voltage: 5V
- Max Power: 25W

3.4× BO Gear Motors (150–200 RPM)

- Operating Voltage: 3V–9V
- Speed: 150–200 RPM
- No-load Current: ~150mA

4.nRF24L01 Wireless Receiver Module



- Operating Voltage: 1.9V–3.6V (3.3V typical)
- Frequency Band: 2.4 GHz ISM
- Range: Up to 100 meters (line of sight)
- Data Rate: Up to 2 Mbps

5.2× 18650 Li-ion Cells

- Type: 18650 Lithium-Ion
- Nominal Voltage: 3.7V (4.2V full, ~3.0V
- empty)
  Capacity: 3200mAh

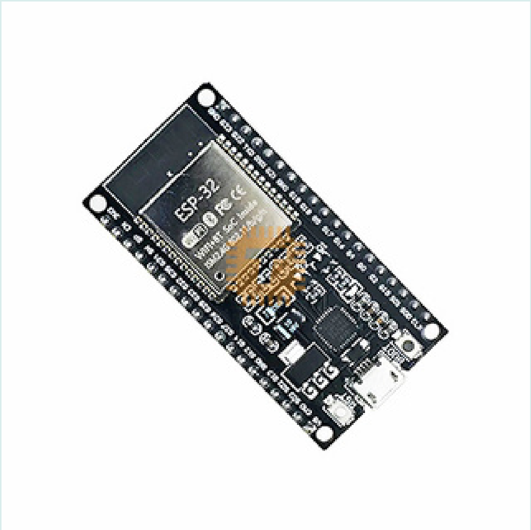
6.Dual-Axis Analog Joysticks

- Operating Voltage: 5V
- Output: Analog (X, Y axes) + Digital (integrated push button)

7.LM2596 Buck Converter Step Down Power Module

- Input Voltage: 2V–24V
- Max Output Current: 2A
- Efficiency: ~93%

8.ESP32 Devkit



- Operating Voltage: 3.3V
- Input Voltage (USB): 5V
- Connectivity: Integrated Wi-Fi
- & Bluetooth
- Core: Dual-Core @ 240 MHz

9.NEMA 17 Stepper Motor

- Step Angle: 1.8° (200 steps per revolution)
- Rated Current: 1.5A–2.0A (model dependent)
- Holding Torque: ~40-45 N·cm

10.A4988 Stepper Motor Driver Module

- Operating Voltage (VMOT): 8V–35V
- Continuous Current: 1A per phase (up to 2A with cooling)
- Microstep Resolution: Full, 1/2, 1/4, 1/8, and 1/16 steps

11.GT2 10mm Timing Belt (PU with Steel Wires) Black Per 1m (MT0436)

- Width: 10 mm
- Pitch: 2 mm
- Material: Polyurethane (PU) with internal Steel Wires
- Tensile Strength: High (reinforced to prevent stretching under load)

12. GT2 20T 20 Teeth Timing Pulley (8mm Bore) 20TB8W6 Silver (MT0072)

- Teeth Count: 20 Teeth (20T)
- Bore Diameter: 8 mm
- Belt Width Compatibility: 6 mm to 10 mm
- Material: Aluminum Alloy
- Pitch: 2 mm (GT2 standard

13.12V 10A SMPS Large Power Supply Metal Casing (PS0019)

- Output Voltage: 12V DC
- Max Output Current: 10A
- Total Rated Power: 120W
- Input Voltage: 110V / 220V AC (Switch Selectable)
- Casing Material: Ventilated Aluminum Metal Mesh Case

14. 12V 2A SMPS Small Power Supply Metal Casing (PS0007)

- Output Voltage: 12V DC
- Max Output Current: 2A
- Total Rated Power: 24W
- Input Voltage: 110V / 220V AC (Switch Selectable)
- Casing Material: Ventilated Aluminum Metal Mesh Case

15. IR Break Beam Sensor 5mm LED (MD0621)

- Operating Voltage: 3.3V to 5V DC
- LED Diameter: 5 mm
- Detection Distance: Up to 50 cm
- Output Type: Digital (High when unblocked, Low when broken)
- Response Time: < 2 ms

16.waterproof LED strip WS2812

- Individually Addressable WS2812B 60LED/M RGB Dream Color LED Pixel Strip 5M
- Model number TS-W2812-60
- LED TYPEE SMD 5050 RGB chip
- Available color RGB

Software Components

- **Platform:**

  HTML, JSON, Python, Arduino IDE, CSS ,JS

- **Main Functions:**

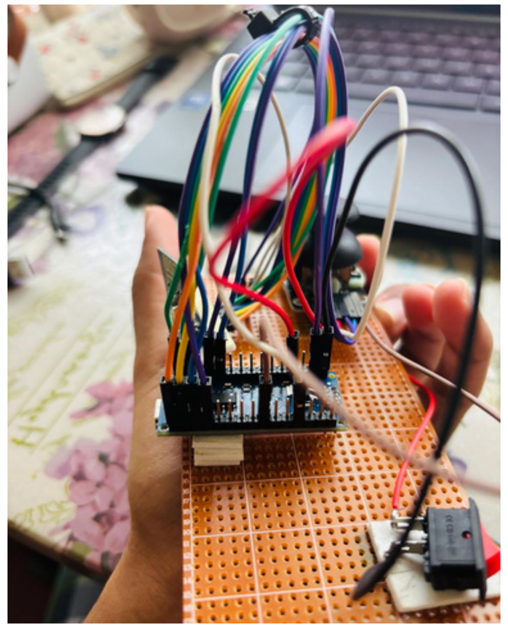
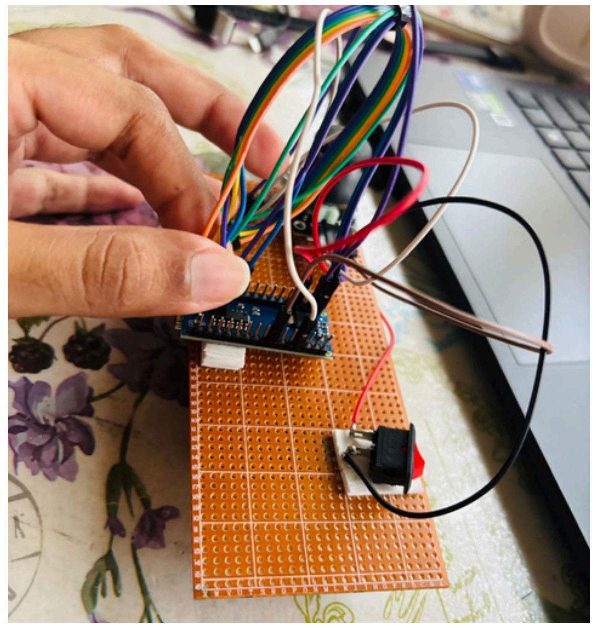
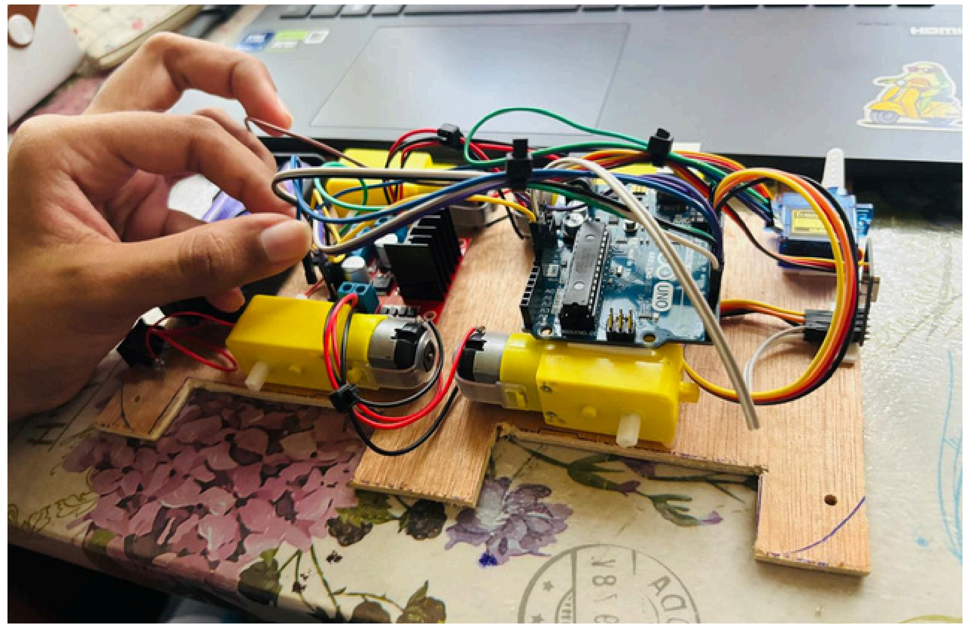
  Display real-time score and timer

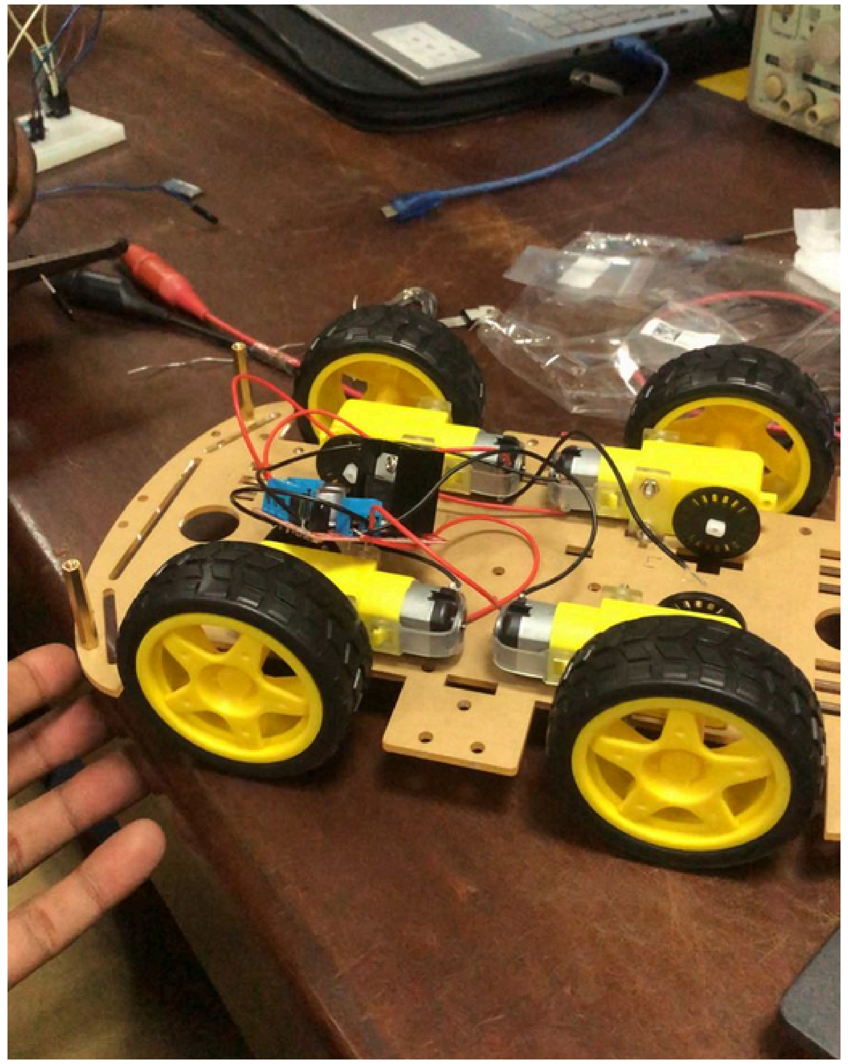
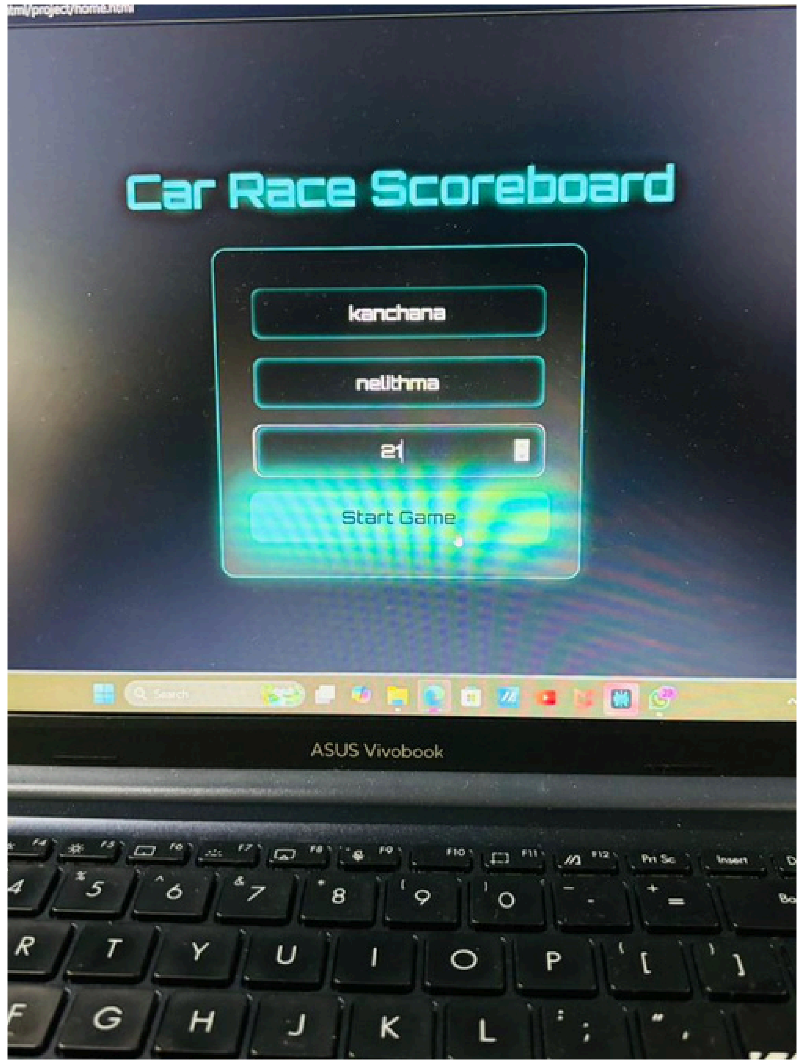
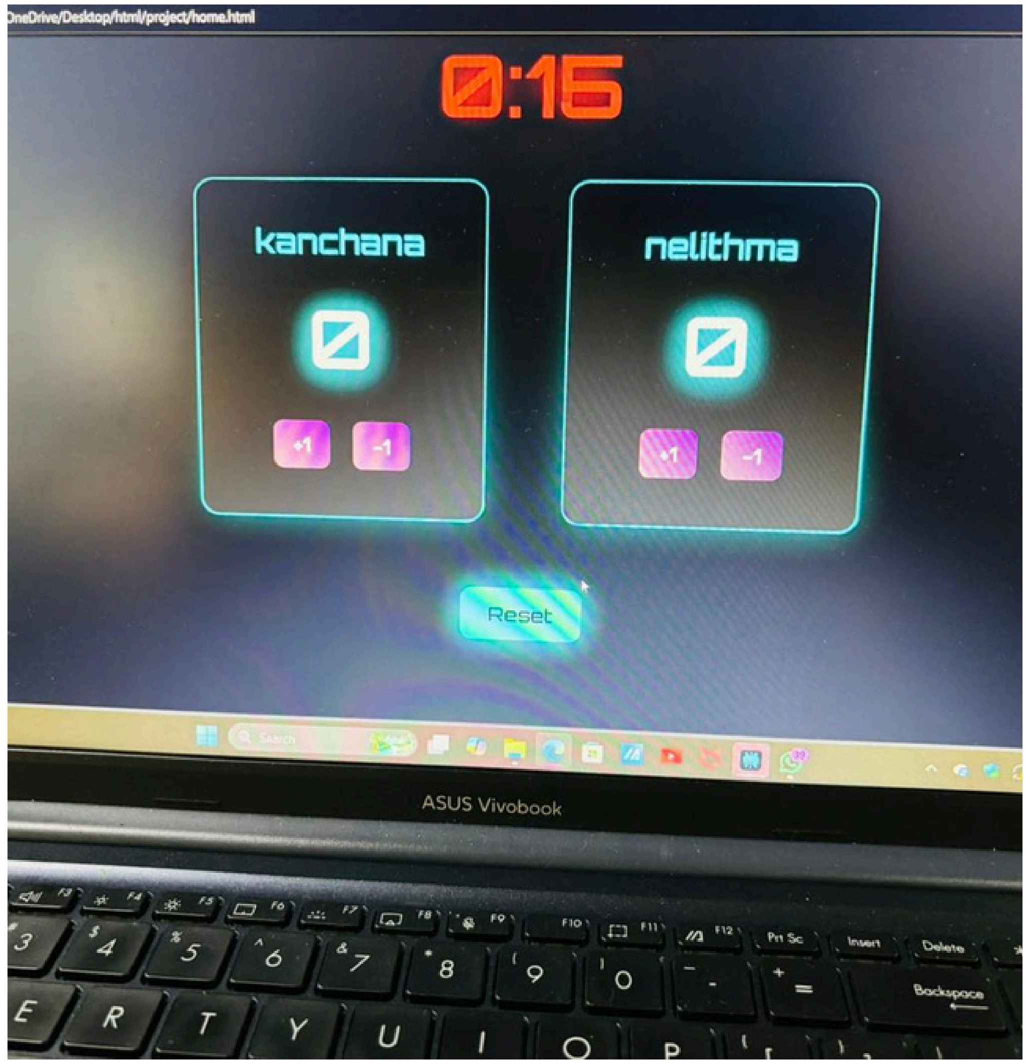
  Detect and register goals via IR break Beam sensor

  Handle game start, pause, and reset

  Communicate goal confirmation

  Display winner at the end of the match




Car Race Scoreboard
kanchana
nelithma
21
Start Game
ASUS Vivobook

OneDrive/Desktop/html/project/home.html
0:15
kanchana
0
+1
-1
nelithma
0
+1
-1
Reset
ASUS Vivobook

WORK DISTRIBUTION

Team member	Role	Responsibility
245031B	Cars design and Game Logic	Arduino Uno Setup for Both cars, Design Chassis, Data Communication through NRF24L01 modules, Initiate Car movements along all directions, implementing the kickers for the cars. Implementing game logics
245056F	Controllers Design and Coding	Arduino Nano setup for Both controllers, Controller setup Design, Data transmission to both car and Arena through NRF24L01 module, Implementing real time update of web application.
245060L	Other arena functions and Coding	Power supply for arena , car and controllers, Implementing other arena outputs (LED, Speaker Module)that change according to game play. App Development: Building App structure and UI.
245057J	PCB design and Goal counting mechanism	Controller PCB Design. Implementing goal counting mechanism for both ends .Connecting ESP32 of arena with web application(ESP32). Overall system testing, debugging and error correction.
245124M	Structure and Goal Keeper mechanism	Creating and design the arena to house all components along with both goal keeper mechanism, design Goal keeper movement (by NEMA 17 motors and rails/belts). Implementing limit switches for goal keepers.

INDIVIDUAL CONTRIBUTION

245031B - P.D.V.D.Gunasekara

1.Configured Arduino Uno boards for both robot cars
(pin assignments, motor drivers, NRF24L01, kicker integration)
2.Designed and assembled robot chassis
(structural stability, durability, and component layout optimization)
3.Assisted in NRF24L01 wireless communication system
(testing and improving low-latency data transmission)
4.Developed omnidirectional movement control
(forward, backward, left, right, and rotational movement tuning)
5.Implemented core gameplay logic
(command handling and system coordination between modules)
6.Supported goalkeeper movement system integration
(motion control planning and testing assistance)
7Led project coordination and team management
(task distribution, progress tracking, testing coordination, troubleshooting)

245056F - K.Lathursan

1. Controller Design and Coding
   - Designed and programmed Arduino Nano–based controllers for both players.
   - Integrated joysticks and buttons for car movement and goalkeeper control.
   - Implemented dual joystick control for responsive gameplay.
2. Wireless Communication
   - Developed NRF24L01 communication between controllers, RC cars, and the arena.
   - Configured multiple wireless nodes for reliable data transmission.
   - Tested and optimized communication stability.
3. Power Management
   - Designed power distribution for the controllers using Li-ion(3.7V 3200mAh) batteries and buck converters.
   - Assisted in arena power management for Arduino, ESP32, sensors, LEDs, and goalkeeper motor. (12V,2A and 12V,5A with buck converters)
   - Ensured stable operation of all subsystems.

4. Assisted Web Application and ESP32 Integration
    - programmed the ESP32 to host a real-time web server and obtained the IP address.
    - Assisted team member in integrating the ESP32 with the developed web application
    - Verified communication between sensors, LEDs, and the web interface.
5. Supported Goalkeeper and Game Logic Support
    - Assisted in implementing goalkeeper control and gameplay functions.
    - Supported NRF24L01 integration for responsive goalkeeper operation.
    - Contributed to system coordination and functionality testing.
6. Collaborated for RC Car Assembly and Integration
    - Assisted in assembling the Arduino Uno–based robot cars.
    - Supported motor driver, NRF24L01 integration.
    - Verified controller-to-car communication and functionality.
7. System Testing and Debugging
    - Integrated controller, arena, and ESP32 subsystems.
    - Performed hardware and software testing and troubleshooting.
    - Improved overall system reliability and gameplay performance.

245060L - S.M.K.Madhushani

1. Power Supply Distribution Network:
   - Two Power Supplies: Used two separate power packs for the arena so the system does not crash.
   - 12V 10A Supply: Gives big power just for the bright stadium lights.
   - 12V 2A Supply: Gives safe, clean power only for the main chips (ESP32 and Arduino Uno).
   - Goalkeeper Motor: Runs on 12V, but uses a Buck Converter to drop it to 5V to run the driver chip safely.
   - Striker Cars: Run completely on their own 18650 batteries.
   - Remote Controllers: Uses a step-up converter to boost battery power to a steady 5V for the Arduino Nano and joysticks.
   - Wireless Module: Receives an exact 3.3V line so the nRF24L01 module works safely without dropping the connection.
2. Interactive Arena Gameplay Outputs:
   - WS2812B LED Strips: Wired addressable lights around the field that change color instantly based on the game status (Green for active play, flashing Red for a goal).
   - Audio System: Programmed the setup to automatically play stadium sound effects and match commentary through the laptop speakers during key events.

3.Web App Architecture & UI/UX Development:

- Clean Scoreboard Webpage: Built a clear, simple dashboard user interface to display live match information to the players.
- Static IP Wi-Fi Connection: Connected the web application wirelessly over local Wi-Fi directly to the unique static IP address of the ESP32 chip.
- Instant Data Transmission: Triggers an immediate data packet sent over the wireless network the exact millisecond the ball crosses the goal line.
- No Webpage Refreshing: Reads the incoming data instantly so the live match timer, score counters, and winner screens update smoothly on the screen without any delay

245057J - B.L.N.N.Liyanage

1) PCB Design

- Designed the complete controller circuitry, integrating dual joysticks, Arduino Nano interface, nRF24L01 wireless communication module, and power management components into a unified schematic.

2)PCB Layout and Fabrication

- Developed the PCB layout, optimized component placement and signal routing, incorporated power regulation circuit , and assembled and tested the fabricated controller boards to ensure reliable operation and wireless communication.

3) Goal Detection and Web Integration Module

- Developed the goal detection system by integrating IR break-beam sensors with the ESP32 microcontroller and web application. Implemented real-time goal detection, data transmission, and automatic score updates on the dashboard, ensuring reliable communication between the hardware and software components.

4) Collaboration with car 2 for integration

- End-to-End Testing: Validated wireless transmissions between the remote control and the Arduino Uno on Car 2 to ensure low-latency movement execution.
- Signal Mapping: Adjusted register parameters to ensure joystick deflection accurately corresponded to the DC motor pulse-width modulation (PWM) rates.

5) Collaboration with goalkeeper

- Positional Syncing: Worked on tracking horizontal joystick data inputs against the linear movement profiles designated for the goalkeeper rail system.

6) Collaboration with Arena

- Fabricated and polished game components to ensure smooth mechanical interactions.

245124M - S.M.M.A.B. Madugalla

1) Build 2 rail and goalkeeper mechanism
   - Linear Slide Fabrication: Constructed the 2-rail linear slide mechanical system providing a precise 20 cm horizontal travel span.
   - Mechanical Drive Assembly: Mounted a NEMA 17 stepper motor coupled with a GT2 timing belt to translate rotational steps into smooth linear defense movements.
2) Goal counting with break beam sensors
   - Optical Tracking: Mounted IR break-beam sensor modules across the goal line entry spaces to automate score logging.
   - Interrupt Configuration: Programmed hardware pin interrupts on the central ESP32 to flag goal states instantly when the optical beam breaks.
3) Install Speakers.
   - Audio System Feedback: Installed and integrated local commentary speakers to trigger real-time audio sound cues upon goal detection.

4) Arena integration

- System Assembly: Assembled the final pitch dimensions (1.5m*5m) alongside structural configurations for the goal boxes.
- Power Distribution: Maintained and structured the 6–9V DC shared power network to reliably feed all microcontrollers, sensors, and actuators inside the arena.

5) Support to designed and assembled robot chassis

Further Improvements

- Improve wireless communication stability and range (more reliable NRF system or upgraded module)
- Enhance robot movement accuracy using better motor control and speed regulation
- Add autonomous features using sensors or camera-based ball tracking
- Improve goal detection using multiple sensors for higher accuracy
- Develop mobile app/web dashboard for control and live match monitoring
- Introduce battery monitoring system with low-power alerts
- Upgrade chassis design for better speed, strength, and durability
- Add intelligent goalkeeper with faster response and predictive movement

Budget

Component	Unit Price (Rs.)	Quantity	Total Price (Rs.)
Geared Motors	130	8	1040
Arduino uno	2000	4	8000
Nrf	350	6	2100
L298N motor driver	420	2	840
switch	50	4	200
car 3D	6000	2	120000
Arduino nano	1300	2	2600
Joystick Module	420	4	1680
LM2596 Buck converter	150	4	600

Component	Unit Price (Rs.)	Quantity	Total Price (Rs.)
battery 3.7V	500	10	5000
PCB		2	9000
NEMA 17 stepper motor	2620	2	5240
A4988 steppper motor driver	280	2	520
ESP 32	1500	1	1500
Power supply Unit	1900	1	1900
Goal Keeper 3D	900	2	1800
Rail(Extrusion bar,gantry plate& etc)	7000	2	14000
Melamaine board & cutting			12000
Total Budget			80020

Conclusion

KickBots Arena is an innovative 2-player robotic football system that blends embedded control, robotics, wireless communication, and computer vision into an interactive game experience.
Each player controls a striker robot and its kicker for attacking (drives the ball toward the goal) and a goalkeeper robot for defense (moves along a rail to block goals)
A Web application acts as the main control and display system.

Thank you very much!

PRESENTED BY OHMIES